Eigenvalues and inverse problems

The optimal decay exponent for the conditioning of sign matrices

Difficulty: extreme **Importance:** interesting to the community

Status: Partially resolved

Rating rationale: Extreme reflects a sharp asymptotic rate tied to longstanding gaps in Hadamard constructions; community impact is a quantitative limit on nearly orthogonal sign matrices.

Partial bound — 2026-09-11

Partial bound by Matthew J. Colbrook, Department of Applied Mathematics and Theoretical Physics, University of Cambridge. Complete manuscript · PDF · LaTeX. **Theorems 1–2 and equations (4)–(8).**

The odd-order obstruction proves $\alpha_* \leq 1/2$ by an all-orders Gram-matrix variance argument. Theorem 2 also strengthens the condition-number lower bound for $n \equiv 2 \pmod 4$. Combined with the previously recorded construction bound, the interval is $17/92 \leq \alpha_* \leq 1/2$. The exact value of α_* and a matching exponent construction remain open; this entry continues to count as open.

The complete argument received an independent Codex-agent **PASS** (partial result only) on 11 September 2026. The review report records the exact scope and a hash of the original reviewed manuscript. The mathematical sections remain unchanged in the authored version. Original ChatGPT generation is disclosed; no external human peer review or formal proof certificate is asserted. Submission and verification record.

The difficulty and importance ratings still apply to the surviving exact-exponent question.

The review also checked Alexeev–Jasper–Mixon, version 2, dated 18 August 2026, §5, Problem 11; it retains the earlier construction exponent and obstruction.

Context and notation

For $n \geq 1$, define

$$h(n) = \min_{A \in \{-1,1\}^{n\times n}} \kappa_2(A), \qquad \kappa_2(A) = \frac{\sigma_{\max}(A)}{\sigma_{\min}(A)},$$

with $\kappa_2(A) = +\infty$ for singular A.

Problem statement

Determine the exact number

$$\alpha_* = \sup\{\alpha \geq 0 : \exists C > 0\ \forall n \geq 1,\ h(n) - 1 \leq C n^{-\alpha}\}.$$

The constant C may depend on α, but not on dimension. This specifies the uniform asymptotic power exponent; logarithmic factors do not change the supremum. Orders admitting exact Hadamard matrices have $h(n) - 1 = 0$, which are included without taking logarithms of zero. Before the 2026-09-11 update above, the recorded bounds were $17/92 \leq \alpha_* \leq 1$.

References

Alexeev, Jasper, and Mixon, *Asymptotically optimal approximate Hadamard matrices*, §6, Problem 11; the supremum above is an editorial precise formulation of its decay-exponent question. Steinerberger, *Open Problems*, Problem 69, November 2025 update, identifies the sharp rate as unresolved.

Earlier status check — 2026-09-08

Searches for *approximate Hadamard 17/92 2026* and *approximate Hadamard condition 2026 sharp* found no exact exponent. A global constant as in IS-04 does not specify a decay exponent, while an asymptotic exponent permits finitely many exceptions to any proposed sharp constant. The two independently posed problems are therefore retained separately.

Audit update — 2026-09-10

Rechecked Alexeev–Jasper–Mixon, §6, Problem 11, and Steinerberger's problem collection, Problem 69. Both leave the sharp rate unresolved. Searches for later approximate-Hadamard conditioning exponents found no determination of the optimum; a nonsharp exponent bound does not by itself resolve the target.